

Robust Benchmarking via Multi-Sample Evaluation

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Introduction

Modern model evaluation often relies on a single held-out test set and a point estimate such as accuracy or F1. This practice can be misleading for two reasons. First, one-shot testing confounds the true performance of a model with the idiosyncrasies of a particular sample. Second, extremely high scores (e.g., 99–100%) are frequently artifacts of unrepresentative class balance or leakage rather than genuine generalization. To obtain a more faithful picture, we argue that a benchmark should be constructed with *explicit data composition* and *multi-sample testing* at its core, and that reporting should center on *ranges* and *uncertainty* rather than single numbers.

Data composition and realism. Let $\mathcal{D}$ denote a pool of labeled examples partitioned into positives $\mathcal{D}^+$ and negatives $\mathcal{D}^-$. A realistic benchmark specifies a target class ratio (π^+, π^-) with $\pi^+ + \pi^- = 1$ (e.g., $\pi^+ = 0.8$, $\pi^- = 0.2$ or $\pi^+ = 0.9$, $\pi^- = 0.1$), reflecting the expected operational prevalence. Benchmarks may be constructed per dataset (single-source baseline) or by mixing multiple sources (pooled baseline) to capture heterogeneity.

Repeated out-of-sample evaluation. Given a model f and a metric $m(\cdot)$ (e.g., Accuracy, Precision, Recall, F1, AUROC), we draw R independent *out-of-sample* test sets $\{\mathcal{T}_r\}_{r=1}^R$ from $\mathcal{D}$ according to (π^+, π^-) , each of size n . We compute $m_r = m(f; \mathcal{T}_r)$ for every replicate. The reporting target becomes the sampling distribution summary:

$$\bar{m} = \frac{1}{R} \sum_{r=1}^R m_r, \quad s_m = \sqrt{\frac{1}{R-1} \sum_{r=1}^R (m_r - \bar{m})^2}.$$

Instead of a single point, we present $\bar{m} \pm s_m$ and a $(1 - \alpha)$ confidence interval (CI). For large R , a normal-approximate CI is

$$\text{CI}_{1-\alpha} : \quad \bar{m} \pm z_{1-\alpha/2} \cdot \frac{s_m}{\sqrt{R}},$$

or the Student- t variant when R is modest. This reframes evaluation as *performance with quantified uncertainty*.

Multiple baselines and factorization of variability. Beyond a single composition, we consider a grid of benchmark settings: (i) **per-dataset** baselines (each source independently), (ii) **mixed** baselines (sources pooled at controlled proportions), and (iii) **composition sweeps** across $\pi^+ \in \{0.5, 0.8, 0.9\}$ (and $\pi^- = 1 - \pi^+$). For each setting, we repeat the sampling protocol to estimate $(\bar{m}, s_m, \text{CI})$. This exposes sensitivity to dataset origin and class balance, two dominant axes of real-world variation.

Comparing multiple models. Let $\mathcal{F} = \{f^{(1)}, \dots, f^{(K)}\}$ denote competing models (e.g., distinct LMs or systems). We evaluate each $f^{(k)}$ under the same sampling plan and report the collection $\{(\bar{m}^{(k)}, s_m^{(k)}, \text{CI}^{(k)})\}_{k=1}^K$ per benchmark setting. Pairwise differences can be tested by resampling-aware procedures. For example, if $m_r^{(a)}$ and $m_r^{(b)}$ are paired by identical $\mathcal{T}_r$, one may use a paired t -test on $\Delta_r = m_r^{(a)} - m_r^{(b)}$, reporting

$$\bar{\Delta} = \frac{1}{R} \sum_{r=1}^R \Delta_r, \quad s_{\Delta} = \sqrt{\frac{1}{R-1} \sum_{r=1}^R (\Delta_r - \bar{\Delta})^2},$$

and a CI for $\bar{\Delta}$. When distributional assumptions are doubtful, a nonparametric alternative (e.g., permutation or bootstrap) provides robust p -values and CIs.

From single numbers to ranges and decisions. Operational decisions rarely hinge on a single-point score; they trade off risk and benefit under uncertainty. Therefore, we encourage reporting *performance bands* (e.g., violin/interval plots over replicates), together with the estimated probability that one model outperforms another by a practically meaningful margin (e.g., $\Pr[\bar{\Delta} > \delta]$ for a chosen effect size δ). This connects statistical significance to *practical significance*.

Controlling for synthetic and “peaky” cases. Where positive or negative examples can be curated or synthesized, we incorporate stress sets that are intentionally “hard” (e.g., distributional shifts, class-ambiguous or high-confusability items). These stress sets are included as additional benchmark cells, evaluated with the same replicate protocol, to surface brittleness that average-case metrics may hide.

Recommended reporting template (drop-in). For each benchmark cell (dataset source, class ratio, stress/regular), report:

1. Replicate count R and test size n ; sampling recipe (with/without replacement).
2. Class composition (π^+, π^-) and any mixing proportions across sources.
3. Metric(s) m with $\bar{m}$, s_m , and $(1 - \alpha)$ CI.
4. Model comparisons: $\bar{\Delta}$, CI, and p -value (paired if applicable).
5. Visual summary: interval/violin plots across replicates; optionally calibration or PR/ROC curves aggregated over replicates.

Notation cheat sheet (for quick reuse).

- $\mathcal{D}$: labeled pool; $\mathcal{D}^+$ positives, $\mathcal{D}^-$ negatives.
- (π^+, π^-) : target class ratio; n : test size; R : replicate count.
- $\mathcal{T}_r$: r -th out-of-sample test set; m_r : metric on $\mathcal{T}_r$.
- $\bar{m}, s_m$: replicate mean and standard deviation; CI: confidence interval.
- $f^{(k)}$: k -th model; Δ_r : replicate-wise paired difference between two models.

Why this matters. This protocol converts evaluation from a one-time number into a distributional characterization that respects data realism (class balance), assesses stability (replicates), and supports principled comparison (CIs, p -values, and effect sizes). It reduces the risk of overclaiming from flattering splits and provides actionable uncertainty that downstream stakeholders can plan against.